

ELECTRIC CIRCUIT ANALYSIS

The Ultimate Comprehensive Master Lecture Notes

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Target Audience: Undergraduates, JEE Advanced, GATE, and Professional Engineers

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1. Fundamental Circuit Quantities & SI Units

To understand the formulation of electrical networks, one must strictly establish and rigorously define the basic quantities of electrical transport systems.

Electric Charge (q or Q): Charge is an intrinsic property of matter. The fundamental unit is the Coulomb (C). One electron carries roughly -1.602×10^{-19} C of charge. All electrical current manifests as the structured net flow of these charge carriers.

Electric Current (i or I): Current is defined as the time-differential rate of charge passage through an arbitrary bounded cross-sectional plane. Mathematically: $i(t) = dq(t)/dt$. The basic SI unit is the Ampere (A), equivalent to one Coulomb per second (C/s). By absolute convention, the direction of current is designated as the direction of net positive charge flow.

Voltage / Electric Potential Difference (v or V): Voltage represents the spatial line-integral of the electric field between two points, denoting the work done per unit charge required to relocate a test charge from a reference node to a specific node. Mathematically: $v = dw/dq$. The standard unit is the Volt (V), equivalent to Joules per Coulomb (J/C).

Electric Power (p or P): Power represents the instantaneous differential rate at which energy is delivered to or drawn from an item. Mathematically: $p(t) = dw/dt = (dw/dq) * (dq/dt) = v(t) * i(t)$. Power is quantified in Watts (W), where $1 \text{ W} = 1 \text{ J/s}$.

Passive Sign Convention (PSC): PSC states that if current enters the positive terminal of an element, the power $p = v * i$ represents power absorbed by that element. Conversely, if current exits the positive terminal, $p = v * i$ represents power delivered by the element.

2. Core Circuit Components (R, L, C, V, I)

Electrical components are categorized as either passive (dissipate or store energy) or active (generate energy). Let us review the fundamental passive and active models.

Passive Circuit Elements

- 1. Resistor (R):** An element that models energy dissipation. The mathematical relation is governed exclusively by Ohm's Law: $v(t) = R * i(t)$. The unit is the Ohm (Ω). Conductance (G) is the reciprocal of resistance: $G = 1/R$, measured in Siemens (S).
- 2. Inductor (L):** An element that stores energy within an induced magnetic field when current passes through it. The voltage drop across an inductor is proportional to the time rate of change of the current: $v(t) = L * di(t)/dt$. The unit is the Henry (H). In a strict DC steady-state scenario ($di/dt = 0$), an inductor acts as an absolute short circuit.
- 3. Capacitor (C):** An element that stores energy inside an electrostatic field generated between two decoupled conductive plates. The current passing through a capacitor is proportional to the time rate of change of the voltage across it: $i(t) = C * dv(t)/dt$. The unit is the Farad (F). In a strict DC steady-state scenario ($dv/dt = 0$), a capacitor acts as an absolute open circuit.

Active Circuit Elements (Sources)

- 1. Independent Voltage Source:** Maintains a specified terminal voltage completely independent of the output current demanded by the external network.
- 2. Independent Current Source:** Maintains a specified terminal current completely independent of the output voltage developed across its terminals.
- 3. Dependent Sources:** Sources whose output value is linearly controlled by a separate voltage or current elsewhere in the circuit network. These are divided into four explicit structural models:
 - Voltage-Controlled Voltage Source (VCVS): $V_{out} = A * V_{control}$
 - Current-Controlled Voltage Source (CCVS): $V_{out} = H * I_{control}$
 - Voltage-Controlled Current Source (VCCS): $I_{out} = G * V_{control}$
 - Current-Controlled Current Source (CCCS): $I_{out} = B * I_{control}$

3. Fundamental Governing Laws (Ohm's, KCL, KVL)

The entire framework of circuit analysis is built upon three foundational pillars: Ohm's Law, Kirchhoff's Current Law (KCL), and Kirchhoff's Voltage Law (KVL).

Ohm's Law & Physical Intuition

Ohm's Law states that for a constant temperature, the current passing through a linear resistor is directly proportional to the potential difference across it. R is the constant of proportionality. Mathematically: $V = I * R$. If a material violates this linear ratio, it is non-ohmic. Power dissipated in an ohmic resistor is formulated as: $P = V * I = I^2 * R = V^2 / R$. Resistors can only absorb energy, converting it permanently into thermal heat.

Kirchhoff's Current Law (KCL)

KCL is an algebraic manifestation of the **Law of Conservation of Charge**. It asserts that charge cannot pool or accumulate at any isolated structural point within a circuit. Therefore, the algebraic sum of all electrical currents entering any independent node must equal zero.

Mathematically expressed: $\text{Sum}(I_{\text{in}}) = \text{Sum}(I_{\text{out}})$ or $\text{Sum}_{k=1}^n I_k = 0$. When writing KCL equations, entering currents are typically assigned a positive sign, and exiting currents are assigned a negative sign (or vice versa, provided consistency is maintained). KCL applies universally to any enclosed closed boundary, often referred to as a supernode.

Kirchhoff's Voltage Law (KVL)

KVL is an algebraic manifestation of the **Law of Conservation of Energy**. It asserts that the total energy gained by a unit charge traversing a closed loop must equal the total energy lost across the passive components within that same loop. Therefore, the algebraic sum of all potential drops and potential rises around any closed path must equal zero.

Mathematically expressed: $\text{Sum}_{k=1}^m V_k = 0$. When traversing a loop, if you enter the positive terminal of an element, you count it as a voltage drop (+V). If you enter the negative terminal, you count it as a voltage rise (-V). KVL is independent of the nature of the circuit elements (linear, non-linear, passive, active).

4. Network Topologies & Simplification Techniques

To analyze a complex circuit efficiently, one must first recognize its structural topological connections and apply network reduction techniques to minimize the number of equations.

Topological Definitions

Branch: Represents a single two-terminal element (such as a resistor, source, or capacitor).

Node: An explicit electrical point of connection where two or more distinct branches meet.

Loop: Any closed independent path formed by starting at a node, passing through a sequence of connected branches, and returning to the starting node without passing through any node twice.

Mesh: A special subset of a loop that does not enclose any internal loops within its boundaries.

Network fundamental relation: $b = l + n - 1$, where b is branches, l is independent loops, and n is nodes.

Series and Parallel Combinations

Elements are in **series** if they are chained sequentially and share the exact same current. For series resistors: $R_{eq} = R_1 + R_2 + \dots + R_n$.

Elements are in **parallel** if they are connected directly to the same two nodes and experience the exact same voltage drop. For parallel resistors: $1/R_{eq} = 1/R_1 + 1/R_2 + \dots + 1/R_n$.

Delta-Wye (Δ -Y) Transformations

When resistors form a configuration that is neither in pure series nor in pure parallel, a Delta-Wye transformation is required. A Delta (Δ) network consists of three resistors connected in a closed ring loop (R_{ab} , R_{bc} , R_{ca}). A Wye (Y) network consists of three resistors connected to a common central node (R_1 , R_2 , R_3).

Delta to Wye Formulas:

$$R_1 = (R_{ab} * R_{ca}) / (R_{ab} + R_{bc} + R_{ca})$$

$$R_2 = (R_{ab} * R_{bc}) / (R_{ab} + R_{bc} + R_{ca})$$

$$R_3 = (R_{bc} * R_{ca}) / (R_{ab} + R_{bc} + R_{ca})$$

Wye to Delta Formulas:

$$R_{ab} = (R_1 * R_2 + R_2 * R_3 + R_3 * R_1) / R_3$$

$$R_{bc} = (R_1 * R_2 + R_2 * R_3 + R_3 * R_1) / R_1$$

$$R_{ca} = (R_1 * R_2 + R_2 * R_3 + R_3 * R_1) / R_2$$

5. Systematic Node-Voltage Analysis Method

Nodal analysis provides a universal, highly organized algorithm to solve any electrical network by using node voltages as the primary independent variables.

Algorithmic Execution Steps

1. Select an arbitrary node as the system **Reference Node** (Ground), assigning it a potential of exactly 0V.
2. Label all remaining non-reference nodes with unknown independent voltage variables ($V_1, V_2, \dots V_{n-1}$).
3. Apply Kirchhoff's Current Law (KCL) to every single non-reference node. Express branch currents in terms of node voltages using Ohm's Law: $I_{\text{branch}} = (V_{\text{source_node}} - V_{\text{destination_node}}) / R$.
4. Solve the resulting system of linear simultaneous equations to determine all unknown node voltages.

The Concept of a Supernode

If an independent or dependent voltage source is connected directly between two distinct non-reference nodes, applying KCL directly becomes difficult because the current through a voltage source is completely unconstrained. To resolve this, the two nodes and the voltage source connecting them are enclosed inside a single generalized boundary called a **Supernode**.

You write a single combined KCL equation for the entire supernode boundary. To obtain the additional equation needed to solve for both unknown voltages, you formulate a constraint equation dictated by the voltage source: $V_{\text{positive_node}} - V_{\text{negative_node}} = V_{\text{source_value}}$.

Formal Nodal Matrix Representation

For a circuit without dependent sources or supernodes, the system can be written as $[G][V] = [I]$, where $[G]$ is the conductance matrix, $[V]$ is the node voltage vector, and $[I]$ is the net independent current source vector entering each node.

6. Systematic Mesh-Current Analysis Method

Mesh analysis is the structural dual method to nodal analysis. It uses imaginary loop currents, known as mesh currents, as the independent variables to solve a circuit.

Algorithmic Execution Steps

1. Verify that the circuit is strictly **planar** (meaning it can be drawn on a flat sheet of paper without any crossing branches). Mesh analysis cannot be applied to non-planar topologies.
2. Identify all distinct meshes (loops that enclose no internal loops). Assign an imaginary circulating current to each mesh ($I_1, I_2, \dots, I_m$), conventionally drawn in a clockwise direction.
3. Apply Kirchhoff's Voltage Law (KVL) around each mesh. Express individual resistor voltage drops by taking the net current passing through them: $V_{\text{resistor}} = R * (I_{\text{essential_mesh}} - I_{\text{adjacent_mesh}})$.
4. Solve the resulting simultaneous algebraic equations to find all mesh currents.

The Concept of a Supermesh

If an independent or dependent current source exists on a shared branch between two adjacent meshes, writing a KVL equation for those individual meshes is difficult because the voltage drop across a current source is unknown. To solve this, you remove the branch containing the current source to form a combined loop called a **Supermesh**.

You write a single KVL equation around the outer boundary of this supermesh. Then, you write an explicit current constraint equation for the shared branch: $I_{\text{clockwise_direction}} - I_{\text{counter_direction}} = I_{\text{source_value}}$.

Formal Mesh Matrix Representation

For a circuit without dependent sources or supermeshes, the system can be written as $[R][I] = [V]$, where $[R]$ is the resistance matrix, $[I]$ is the unknown mesh current vector, and $[V]$ is the net driving voltage source vector around each mesh.

7. Fundamental Circuit Theorems (Thevenin & Norton)

Network theorems are vital tools that allow engineers to simplify large, complicated circuits into elementary equivalent circuits, focusing analysis on a specific load element.

Thevenin's Theorem

Thevenin's Theorem states that any linear, two-terminal electrical network containing independent sources, dependent sources, and resistors can be completely replaced at its output terminals (A and B) by an equivalent circuit consisting of a single independent voltage source (V_{th}) connected in series with a single resistor (R_{th}).

V_{th} (Thevenin Voltage): The open-circuit voltage measured or calculated across the terminals A and B when the external load is disconnected.

R_{th} (Thevenin Resistance): The equivalent input resistance looking into terminals A and B with all independent sources deactivated (voltage sources replaced by short circuits, current sources replaced by open circuits). If dependent sources are present, R_{th} must be found by applying a test source (V_{test}) at the terminals and computing $R_{th} = V_{test} / I_{test}$.

Norton's Theorem

Norton's Theorem is the dual of Thevenin's Theorem. It states that the same linear network can be replaced at its terminals by an equivalent circuit consisting of a single independent current source (I_n) in parallel with a single resistor (R_n).

I_n (Norton Current): The short-circuit current that flows when terminals A and B are connected together with an ideal short circuit.

R_n (Norton Resistance): The equivalent resistance looking into the terminals, which is identical to the Thevenin resistance ($R_n = R_{th}$).

Source Transformation: Thevenin and Norton circuits are related by: $V_{th} = I_n * R_{th}$.

8. Advanced Circuit Theorems & Maximum Power Transfer

Building upon Thevenin's and Norton's discoveries, we explore additional linear theorems used for multi-source systems and power optimization.

Superposition Theorem

The Superposition Theorem states that in any linear circuit containing multiple independent sources, the net voltage drop across or current through any element is equal to the algebraic sum of the independent voltages or currents produced by each source acting alone.

To isolate an individual source, all other independent sources must be deactivated. Replace independent voltage sources with short circuits (0V) and independent current sources with open circuits (0A). *Important Note:* Dependent sources are never deactivated; they must remain fully active during every step of the superposition process because they are controlled by network variables.

Maximum Power Transfer Theorem

In many engineering applications, the goal is to transfer the maximum possible power from a driving source network to an adjustable external load resistor (R_L). The Maximum Power Transfer Theorem states that maximum power is delivered to the load when the load resistance is exactly equal to the Thevenin equivalent resistance of the source network looking back from the load terminals.

Mathematically: $R_L = R_{th}$. When this condition is met, the maximum power dissipated in the load resistor is calculated as: $P_{max} = (V_{th})^2 / (4 * R_{th})$. Under this condition, the overall system electrical efficiency is exactly 50%, as an equal amount of power is dissipated inside the source network's internal resistance.

9. AC Sinusoidal Steady-State Analysis & Phasors

When a circuit is driven by alternating sinusoidal sources (AC), time-domain differential equations become tedious. We use the phasor transform to convert these into linear algebraic equations.

The Phasor Transform & Euler's Identity

A time-domain sinusoidal signal $v(t) = V_m \cos(\omega t + \phi)$ can be represented using Euler's identity as the real part of a complex exponential: $v(t) = \text{Real}\{ V_m e^{j\phi} e^{j\omega t} \}$. The **Phasor Representation** drops the time-dependent term $e^{j\omega t}$ and retains only the magnitude and phase angle: $V = V_m \angle \phi$.

Complex Impedance (Z)

In the phasor domain, the ratio of phasor voltage to phasor current defines complex **Impedance (Z)**, measured in Ohms (Ω). It generalizes Ohm's law to AC systems: $V = I \cdot Z$.

Component	Time Domain	Phasor Impedance (Z)	Phase Shift
Resistor (R)	$v = R \cdot i$	$Z = R$	0 degrees (In Phase)
Inductor (L)	$v = L \cdot di/dt$	$Z = j \cdot \omega \cdot L$	+90 degrees (Current Lags)
Capacitor (C)	$i = C \cdot dv/dt$	$Z = 1 / (j \cdot \omega \cdot C)$	-90 degrees (Current Leads)

By transforming a circuit into the phasor domain, all standard DC techniques—including KCL, KVL, Nodal Analysis, Mesh Analysis, Thevenin's Theorem, and Norton's Theorem—remain fully valid. You simply replace real numbers with complex numbers.